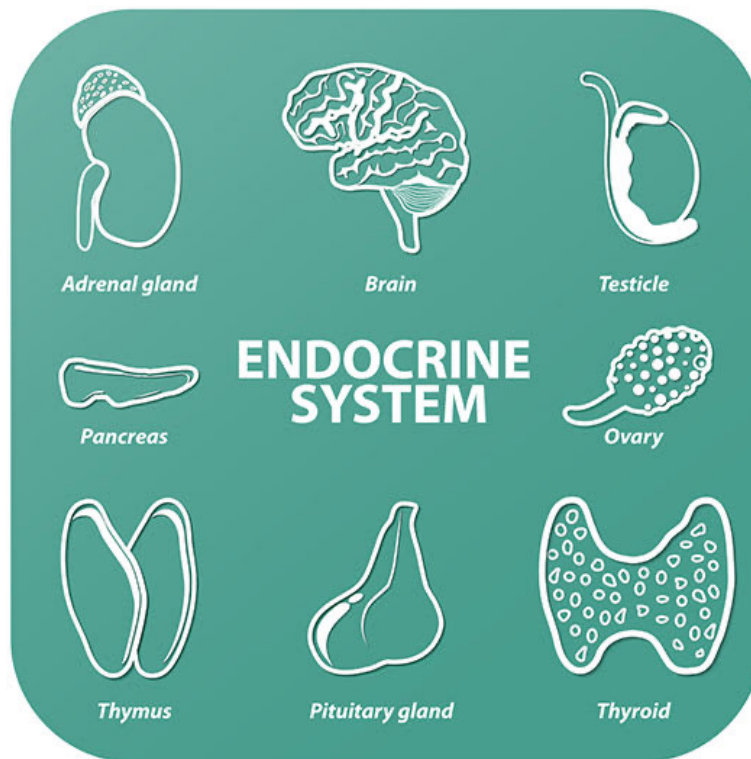
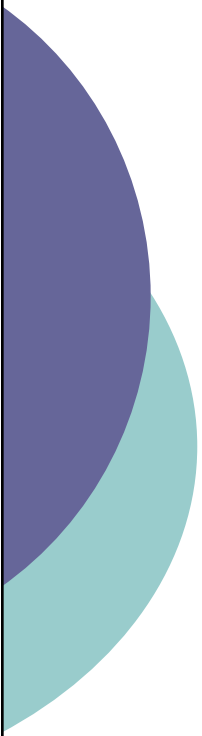


Endocrine System

Instructor – Jeff Schinske



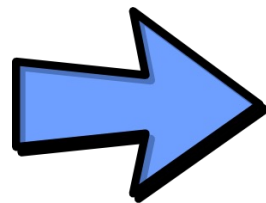


Bio 40C Proposed Norms

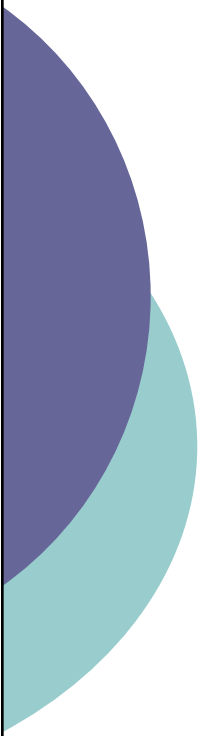
- Seek to enjoy your colleagues and your experiences in class!
- Actively select language that is inclusive, respectful, and welcoming
- Assume good intentions...no heroes and no villains (in this class)
- Recognize that we all have strengths as well as areas for growth
- Be curious, be skeptical
- Ask for help (and expect to receive it!)

Share in your small team...

1. Name
2. Something important to know about you as a person
3. What plans (or costumes?) do you have for this evening?



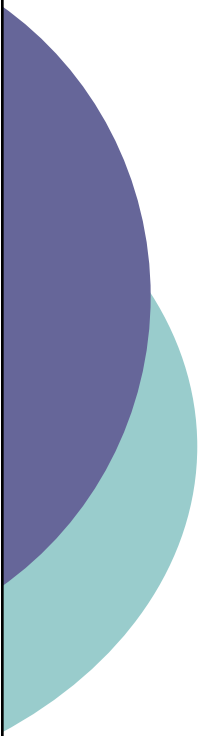
Person wearing lightest colored shirt goes first



Debriefing Last Week

○ End of Class Index Card Reflections

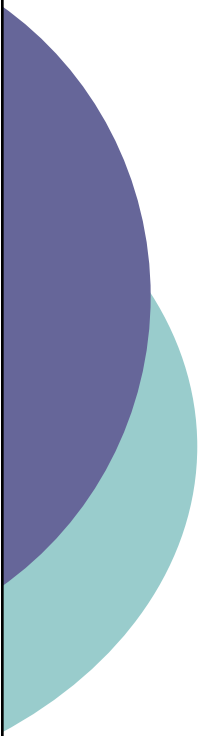
- "It really stood out being able to connect our everyday lives to what we're learning and the effects it has."
- "What stood out most to me was the kidney failure/diabetes part where we diagnosed them and found out treatments."
- "Something that stood out to me is how drinking water can't change your body's pH, other than pee."
- "I loved the lab test! I actually learned things I hadn't before and I left feeling happy and proud."



Debriefing Last Week

○ Last Week's Canvas Work

- "I really enjoyed this section of the class, and I have a much better grasp than I did before on the different compartments of the body and the transport of different solutes and water. I plan on using this for my own personal and professional knowledge as I apply it to future clients. I have been telling my girlfriend everything I learned and I have explained it much more clear than I could have before."
- "I guess now I understand that when we sweat from exercise, we are losing fluids from the extracellular fluid, which causes a shift in the intracellular fluid due to osmosis."
- "I've gained a pretty good "A&P perspective" on these topics as we went over what's happening at the cellular level in great detail. I think I need some more practice with it though to feel completely comfortable."



A Brief Preview of the Week

○ Today

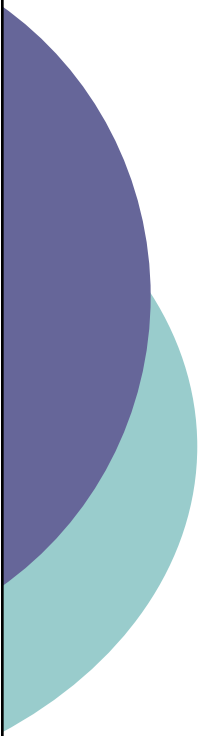
- Getting some practice explaining water and solute balance
- Modeling (with clay!) hormone mechanisms

○ Lab Today

- Water & solute balance

○ Online Lecture Work This Week

- Exploring some “mystery” hormone experiments



What will you be able to do?

After class, you will be able to:

- Explain from an A&P perspective the physiological basis for the consumption of sports drinks.
- Construct models to demonstrate the mechanisms of action for various hormones

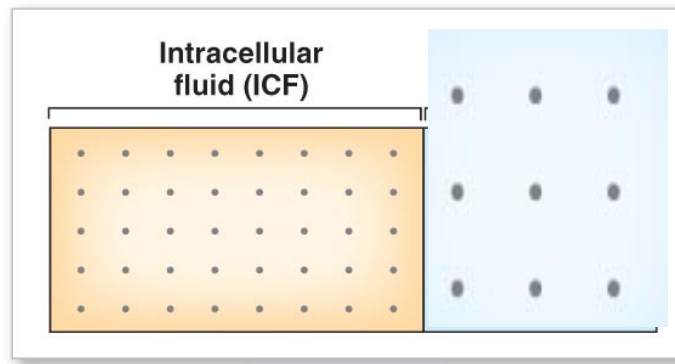
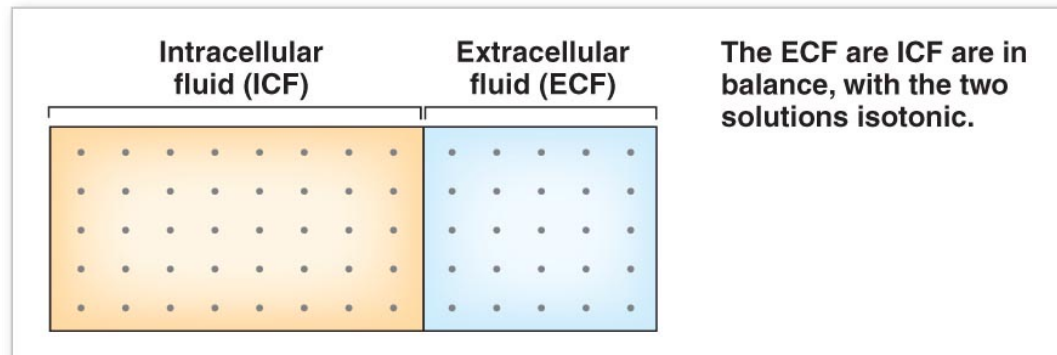
Practicing Our Explanations of the A&P of Water and Solutes

In the video we watched, the University of Florida football players demonstrated a variety of physiological impacts of water imbalances. A couple of the things the players experienced were 1) an intense desire to "guzzle" water, and 2) numerous negative side effects (bloating, inflammation, cramps, etc.) if they did guzzle water.

From an A&P perspective, why did the athletes experience so many negative reactions if they drank tons of water during their intense exercise?



Dehydration, hyponatremia and fluid compartments

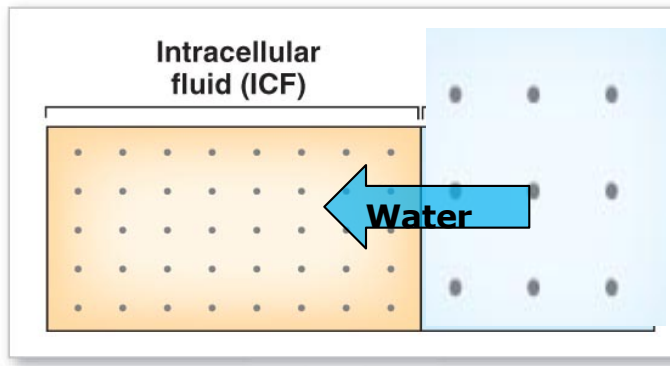


Hyponatremia

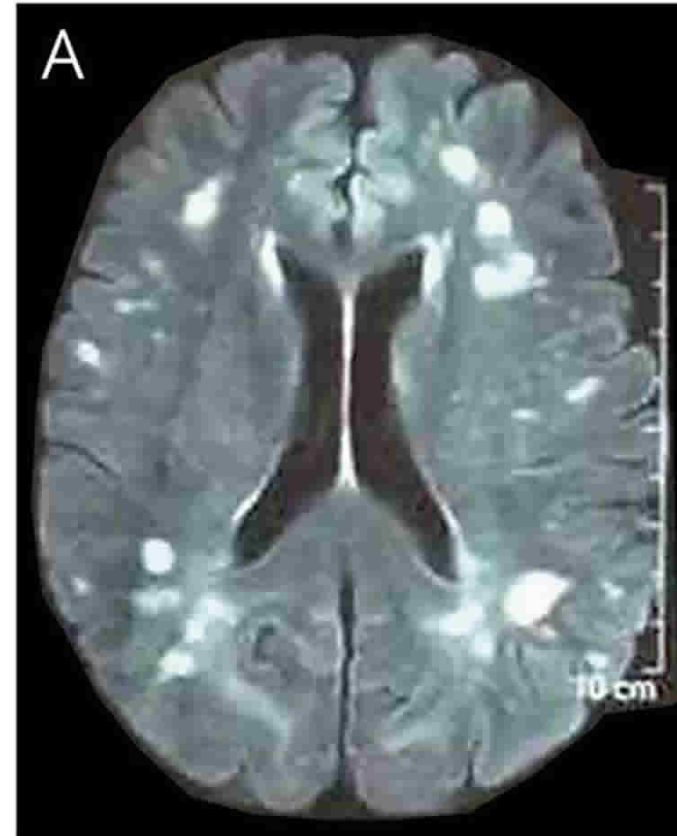
How will this affect water flow between compartments?

How could this lead to headache (and worse) if this happens in the brain?

Net water movement into ICF (into cells)

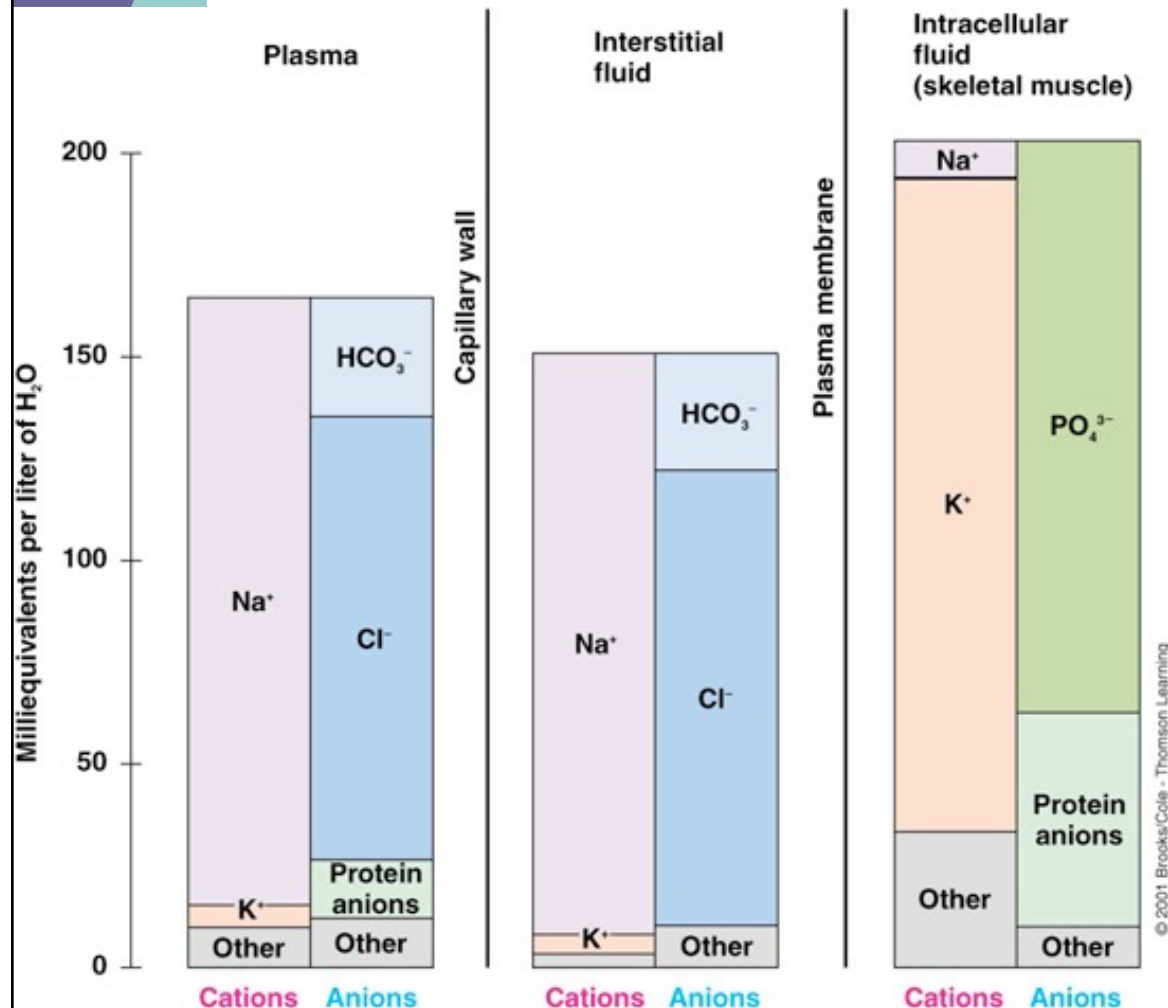


If osmosis causes brain cells to swell, the brain has no room to expand! (trapped in a skull)

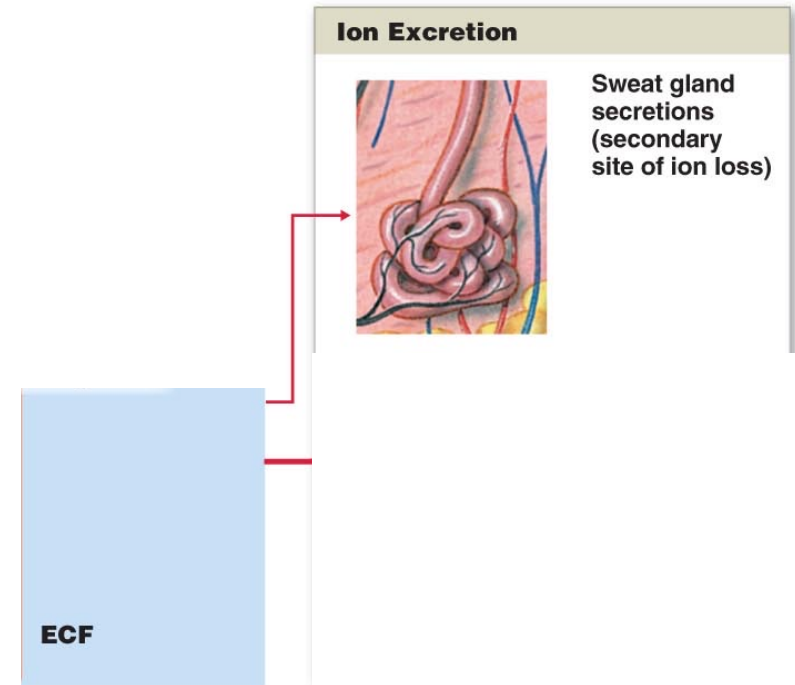


Different electrolytes are found inside and outside cells (due to types of membrane channels, resting potentials)

Mostly Sodium & Chloride in ECF
Mostly Potassium & Phosphate in ICF

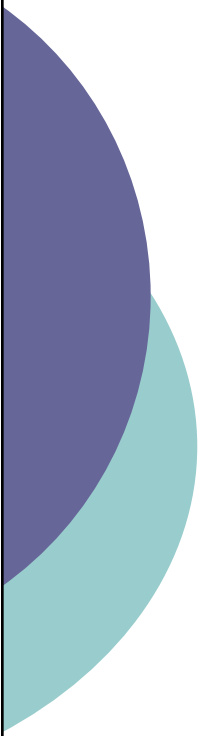


Which fluid compartment does sweat most directly come from?



Which electrolytes do you think football players most lose through sweat?

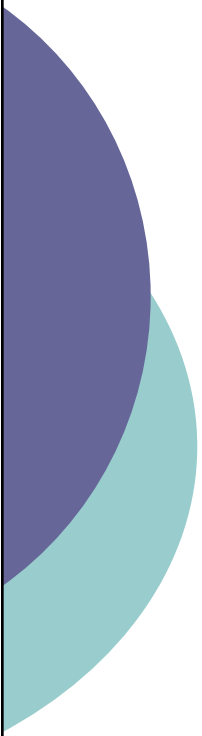
9	Your Name
<i>Your responses...</i>	



What will you be able to do?

After class, you will be able to:

- Explain from an A&P perspective the physiological basis for the consumption of sports drinks.
- Construct models to demonstrate the mechanisms of action for various hormones

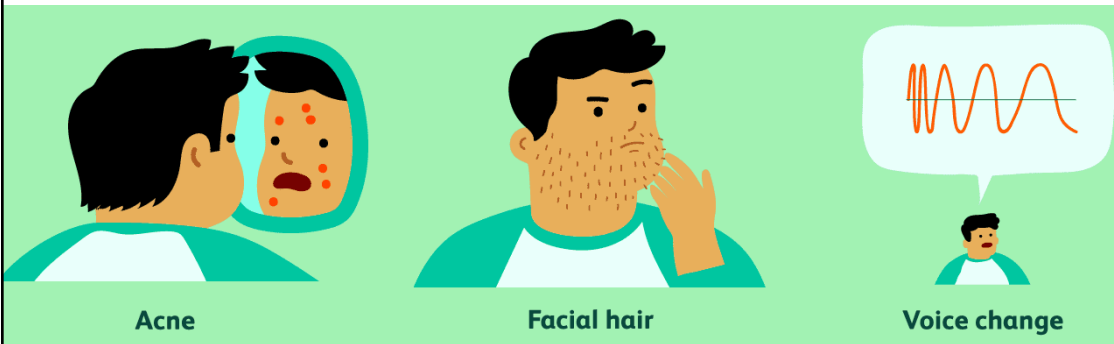
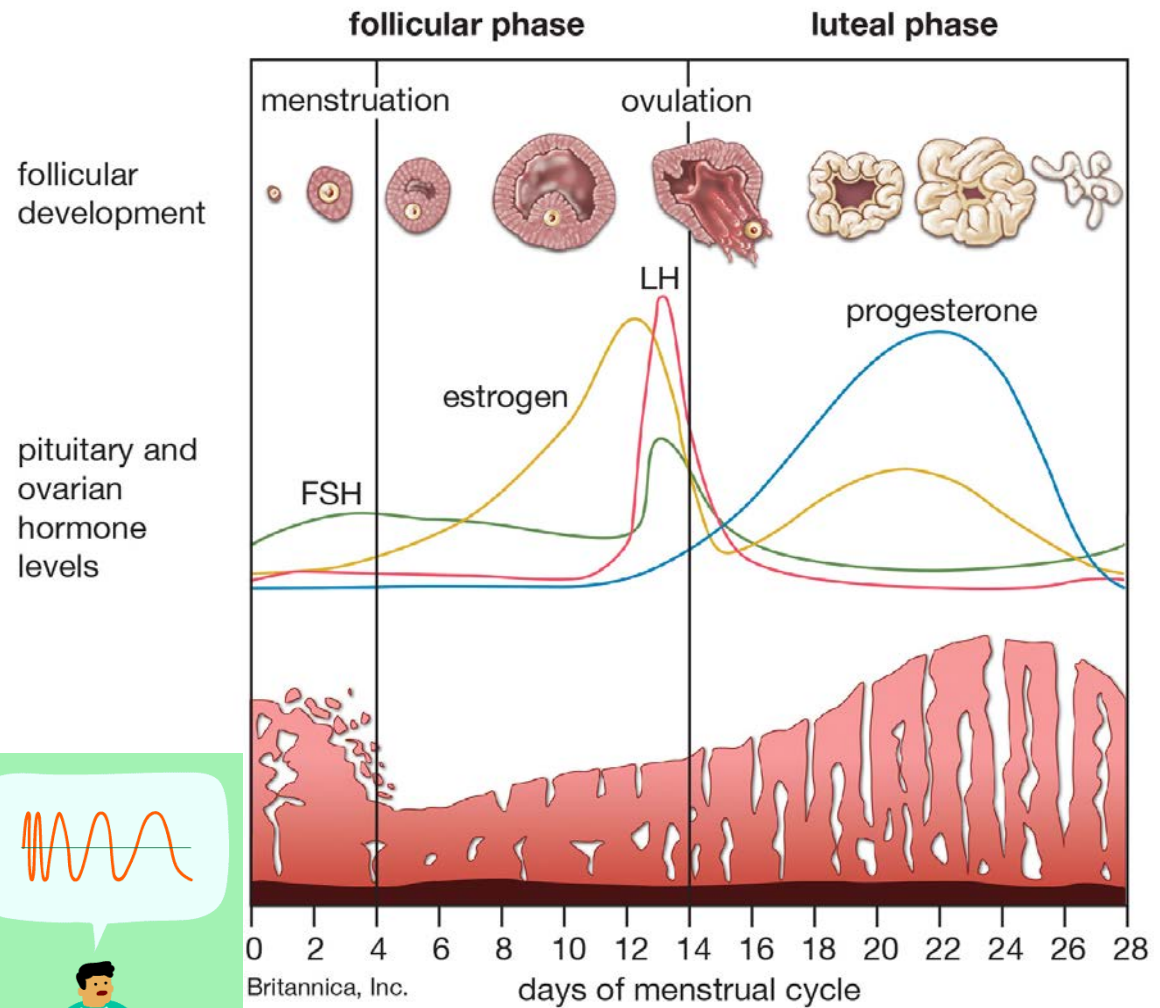


What comes to mind when you
hear the word “hormones?”

Hormones?



The menstrual cycle



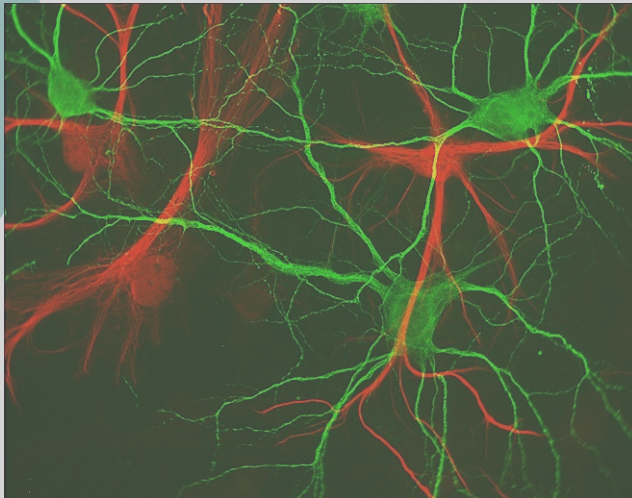
Acne

Facial hair

Voice change

Body's Communication Systems

Nervous System



Neurons

FAST, Short Lasting

**Membrane Potentials
Neurotransmitters**

**Targets = Muscles,
Glands, Neurons**

Endocrine System



Glands & Hormone Secreting Cells

Slower, Longer Lasting

Hormones

**Molecules released in one part of body
that travel and alter activities of cells in
a different part of body**

**Targets = All types of cells (different
for each hormone)**

In a team of 2-3, you will now receive a description of the ways one hormone acts on our cells.



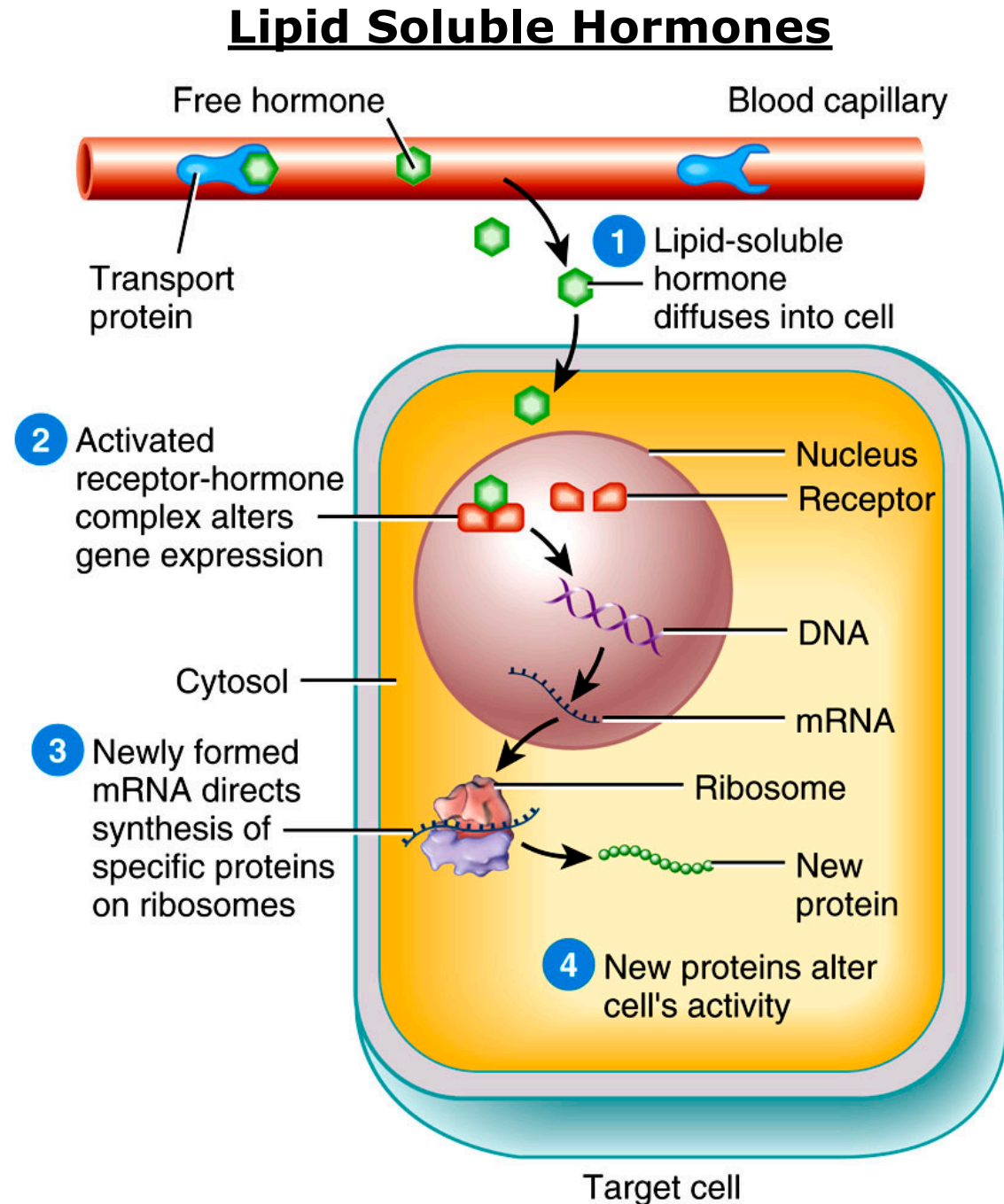
With your partner(s)...

- **Obtain a bag of clay**
- **Review the information about how this hormone acts on cells**
- **Using the information from your paper, create a clay model that shows a cell and the ways the hormone will access and act on the cell.**

Consider...

- **What cell parts do you think are important to show?**
- **What chemicals/molecules do you think are important to show?**

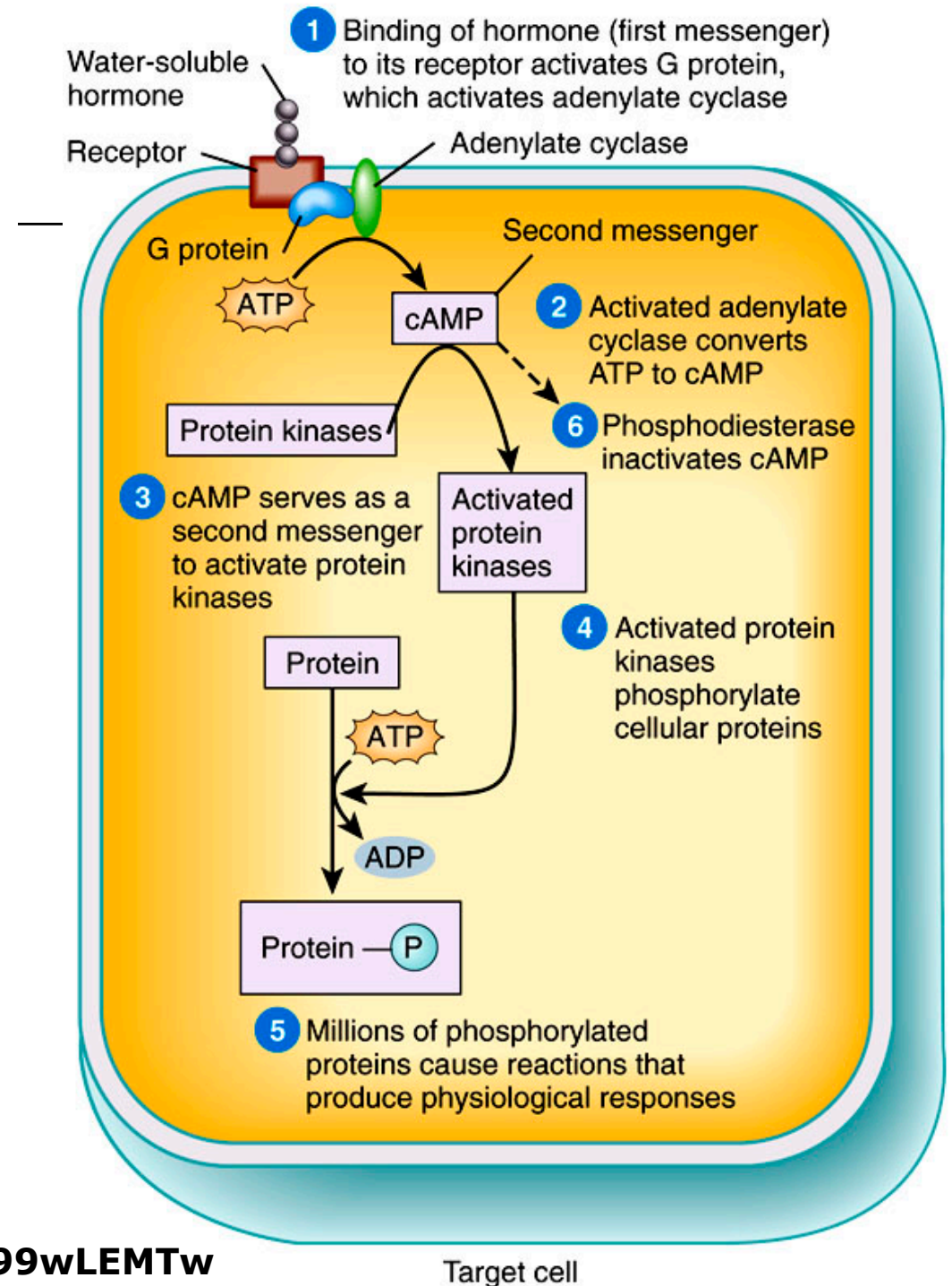
Mechanism of hormone binding and reactions depends on whether hormone dissolves in water or lipids



"Lipid Soluble Hormones (steroid hormones)"
<https://www.youtube.com/watch?v=DHa8JiqFg5Y>

Mechanism of hormone binding and reactions depends on whether hormone dissolves in water or lipids

Water Soluble Hormones



“Action of epinephrine”

<http://www.youtube.com/watch?v=ejq99wLEMTw>

In a team of 2-3, you will now receive a description of the ways one hormone acts on our cells.



With your partner(s)...

- Obtain a bag of clay
- Review the information about how this hormone acts on cells
- Using the information from your paper, create a clay model that shows a cell and the ways the hormone will access and act on the cell.

Consider...

- What cell parts do you think are important to show?
- What chemicals/molecules do you think are important to show?

Cleaning Up

- **Separate the clay and group it by color. You should have four different colors**
- **Seal the four colors of clay in the baggie to return**

Water soluble or lipid soluble hormone? Why?
What molecules and cell parts can you identify in the models?

